

Trade, Offshoring, and Labor Market Adjustment

Labor II — Week 12

Teaching deck draft

Central question and week repositioning

- How do trade and offshoring shocks transmit through workers, firms, sectors, and places?
- Week 12 is a labor-market adjustment lecture, not a generic trade lecture.
- The same trade shock can move wages, employment, unemployment, nonparticipation, sector shares, and welfare on different horizons.

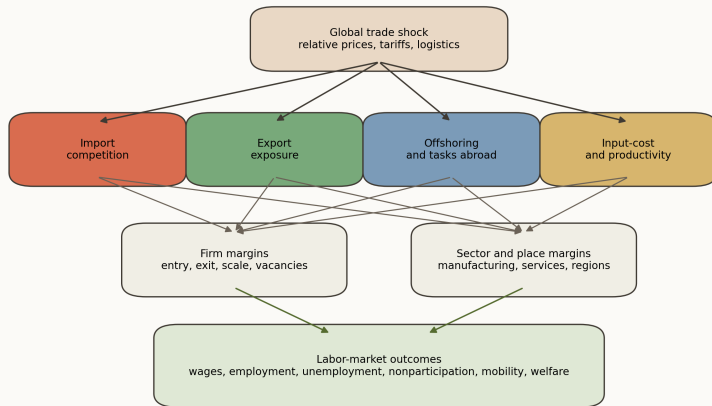
Week 11 to Week 12 bridge

- Week 11: shocks began inside production through task change, automation, and organizational redesign.
- Week 12: shocks begin through global integration, relative prices, and the location of production.
- Both weeks study adjustment, but trade must keep import competition, export exposure, offshoring, and structural change separate.

Why trade is not just a demand shock

- Import competition can contract exposed sectors and places.
- Export access can expand labor demand and wage opportunities.
- Offshoring and imported intermediates can reorganize tasks inside firms.
- Structural change can shift employment from manufacturing to services.
- Mobility frictions determine whether workers, firms, and places adjust together.

Trade channel map



Different trade channels can hit the same labor margin through different propagation paths.

One local labor-market exposure object

$$\text{Exposure}_r = \sum_s \omega_{rs,0} \Delta \text{TradeShock}_s$$

- Baseline sector shares map a sectoral trade shock into local exposure.
- The treatment object changes when the unit is a worker, firm, or place.
- Exposure is not yet a welfare object; it is an incidence object.

What the reduced-form outcome equation identifies

$$\Delta y_r = \beta \text{Exposure}_r + X_r' \Gamma + \varepsilon_r$$

- The observed margin might be employment, wages, unemployment, nonparticipation, or sector shares.
- Local designs are strongest on place-level incidence.
- Migration, worker transitions, and aggregate price adjustment remain partly offstage.

Import competition, export exposure, and offshoring

- Import competition: foreign supply shock in competing sectors.
- Export exposure: market-access gains and local expansion.
- Offshoring: relocation of tasks or intermediate stages abroad.
- Input-cost channel: cheaper intermediates raise productivity for some firms.
- Structural-change channel: trade shifts sector composition and service demand.

China syndrome and local U.S. adjustment

- Autor-Dorn-Hanson: commuting-zone exposure to Chinese import growth.
- Unit: place. Margin: employment, wages, unemployment, nonparticipation.
- Core result: more exposed places saw larger manufacturing decline and weaker labor outcomes.
- Interpretation: identifies place incidence, not full worker welfare or aggregate gains from trade.

Policy exposure and manufacturing decline

- Pierce-Schott links manufacturing decline to the policy environment around China trade integration.
- Unit: industry. Margin: manufacturing employment decline.
- Lesson: trade shocks also operate through uncertainty resolution, firm exit, and industry restructuring.

Dynamic adjustment needs a horizon object

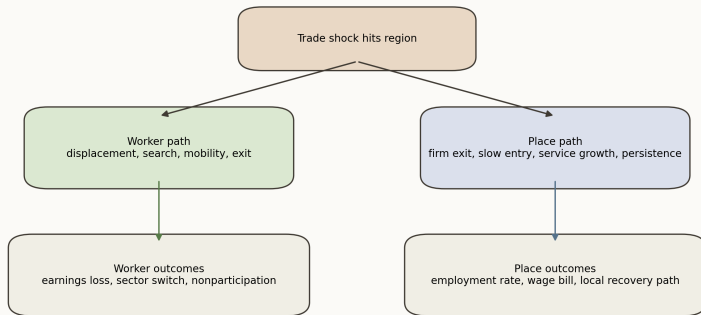
$$y_{rt+h} - y_{rt-1} = \sum_{h \in \mathcal{H}} \beta_h \text{Exposure}_r + X'_{rt} \Gamma_h + \varepsilon_{rt+h}$$

- Short-run displacement and long-run adjustment are different empirical objects.
- Event-time responses discipline claims about persistence and recovery.
- Regional recovery need not imply that initially displaced workers are whole again.

Brazil and dynamic regional adjustment

- Dix-Carneiro-Kovak: regional exposure to tariff reform over long horizons.
- Unit: region over time. Margins: earnings, employment structure, persistent adjustment.
- Main lesson: regional labor markets adjust slowly, so early losses and long-run outcomes can differ sharply.

Worker versus place adjustment



Worker recovery can occur without place recovery, and place recovery can mask persistent worker losses.

Trade-induced structural change

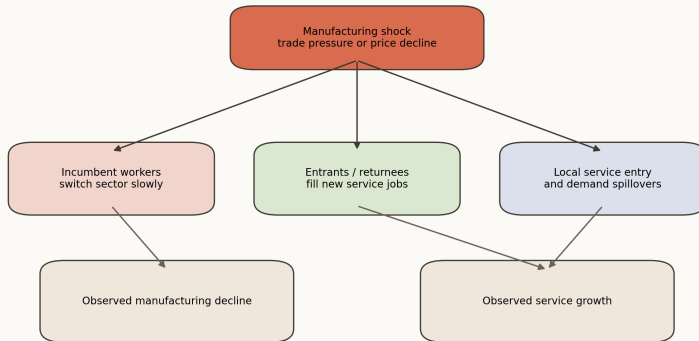
$$\Delta \log \frac{L_M}{L_S} = \eta_p \Delta \log \frac{P_M}{P_S} + \eta_y \Delta \log Y + \eta_a \Delta \log \frac{A_M}{A_S}$$

- Manufacturing-to-service transition can come from relative prices, expenditure shifts, and productivity differences.
- Sectoral reallocation is not enough; the lecture must ask who actually moves.

Germany, structural change, and service transitions

- Dauth-Findeisen-Suedekum: German exposure to Eastern Europe and China.
- Unit: workers and regions. Margins: manufacturing decline, service growth, transition margins.
- Key lesson: observed service expansion is not simply incumbent manufacturing workers gliding into service jobs.

Manufacturing-to-service transitions



Sectoral reallocation can be entry-driven even when incumbent-worker transition is limited.

Offshoring and services trade as a distinct channel

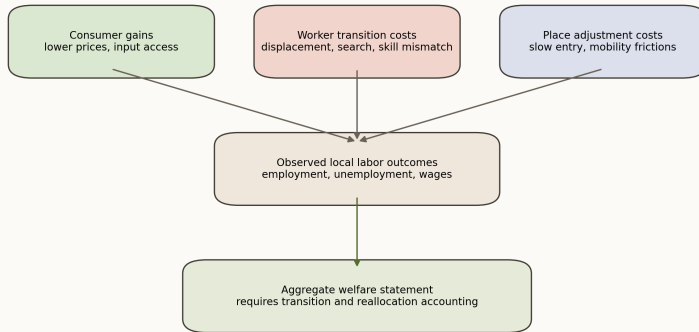
- Feenstra-Hanson: outsourcing changes within-firm organization and relative demand.
- Offshoring is not the same object as final-goods import competition.
- Imported intermediates, fragmented tasks, and services trade can alter skill demand even when local import exposure is modest.

Worker losses, place losses, and welfare

$$\Delta W = \sum_i \pi_i \Delta \text{RealIncome}_i - \sum_i \text{TransitionCost}_i - \sum_r \text{PlaceAdjustmentCost}_r$$

- Worker losses, place losses, and aggregate gains are distinct objects.
- Local labor-market estimates are central for incidence but insufficient for welfare.
- Structured frameworks are needed for policy counterfactuals.

Trade, welfare, and reallocation

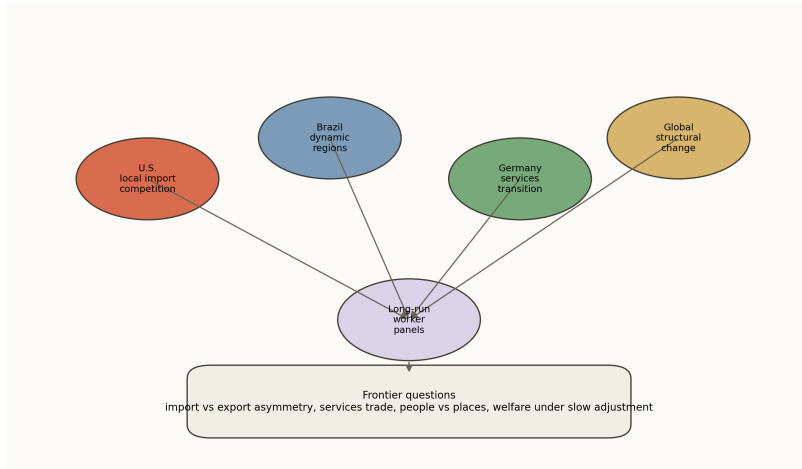


Local incidence is a crucial input, but it is not the whole welfare calculation.

Structured equilibrium frameworks

- Kim-Vogel: trade shocks with unemployment, nonparticipation, and adjustment frictions.
- Caliendo-Dvorkin-Parro: dynamic general equilibrium analysis of the China trade shock.
- What structure adds: transition paths, worker reallocation, consumer gains, and welfare accounting.

Global evidence and frontier directions



New research directions

- Worker panels versus place panels
- Import versus export asymmetry
- Services trade and white-collar offshoring
- Structural change into services and the role of entrants
- Trade, labor-market power, and local concentration
- Welfare and policy evaluation under slow adjustment

Bridge to Week 13 synthesis

- Week 11 and Week 12 together complete the major shock-and-adjustment block of Labor II.
- Week 13 will ask how firms, frictions, institutions, and shocks fit into one research design toolkit.
- The synthesis question is no longer whether trade matters, but which object, margin, and horizon a given design actually identifies.

- Trade affects labor markets through several distinct channels, not just import competition.
- Local exposure designs identify place incidence, not full worker welfare.
- Dynamic evidence and structural change evidence are essential for understanding persistence.
- Worker adjustment, place adjustment, and aggregate welfare should never be conflated.

Appendix: Week 12 lab bridge

- Reproduce: a bounded China-syndrome-style commuting-zone exposure exercise.
- Diagnose: identify the channel, unit, margin, and hidden adjustment margin.
- Transfer: carry the same exposure logic into a synthetic structural-change setting.
- Challenge extension: compare short-run place effects to dynamic regional adjustment.